

6-6c Effects of Inventory Errors on the Financial Statements

Any errors in inventory will affect the balance sheet and income statement. Some reasons that inventory errors may occur include the following:

- Physical inventory on hand was miscounted.
- Costs were incorrectly assigned to inventory using an inventory costing method, such as FIFO, LIFO, or weighted average, that was incorrectly applied.
- Inventory in transit was incorrectly included or excluded from inventory.
- Consigned inventory was incorrectly included or excluded from inventory.

Inventory errors often arise when conducting the end-of-year “physical” inventory. For example, merchandise that was ordered *FOB shipping point* may be in transit at the end of the year and thus, not counted as part of the physical inventory. Even though the inventory has not been received, the title to the merchandise passed to the buyer at the time of shipment and should be included in the buyer’s physical inventory.

Likewise, manufacturers sometimes ship merchandise to retailers who act as the manufacturer’s selling agent. The manufacturer, called the **consignor** ([The manufacturer in a consigned inventory arrangement.](#)), retains title until the goods are sold. Such merchandise, called

consigned inventory ([Merchandise that is shipped by manufacturers to retailers who act as the manufacturer’s selling agent.](#))

, is said to be shipped *on consignment* to the retailer, called the **consignee** ([The retailer in a consigned inventory arrangement.](#)). Any unsold merchandise at year-end is a part of the manufacturer’s (consignor’s) inventory, even though the merchandise is in the hands of the retailer (consignee). At year-end, the retailer (consignee) may incorrectly include the consigned merchandise in its physical inventory or the manufacturer may incorrectly exclude it from its physical inventory.

Errors in the ending inventory affect not only the balance sheet, but also the income statement. In a perpetual inventory system, the end-of-year physical inventory is the basis for the inventory shrinkage adjustment illustrated in [Chapter 5, Accounting for Retail Businesses](#). This adjustment for inventory shrinkage increases Cost of Goods Sold and decreases Inventory. In a periodic inventory system, ending inventory is subtracted from goods available for sale in computing cost of goods sold. In addition, since the ending inventory becomes the beginning inventory of the next period, the income statement of the next period is also misstated.

Pathways Challenge

This Is Accounting!

Economic Activity

Tyson Foods (TSN), one of the world's largest food companies, operates a chicken production process that begins with the hatching of chicks and ends with packaged chicken for sale at your local grocery market. The most desirable chickens are maintained as breeding stock for laying eggs and producing chicks. Once a chick is hatched, it is transported to a growing farm. At the farm, the chicken is housed and fed corn, soybean, and other feed. Once the chicken reaches maturity, it is transported to a processing plant where it is converted into a finished product and packaged for delivery to your local grocery store.

Critical Thinking/Judgment

How should Tyson account for the transportation, boarding, and feed costs related to chickens sent to the processing plant?

Can you think of any other costs related to chickens raised for processing?

Should the costs of the breeder flock be included in inventory?

Exhibit 11 illustrates the effects of inventory errors on the income statement and balance sheet.

Exhibit 11

Effects of Inventory Errors

Ending Inventory Overstated in Current Period			
Current Period		Next Period	
Income Statement		Income Statement	
Sales	Correct	Sales	Correct
Beginning inventory	Correct	Beginning inventory	Overstated
Purchases	Correct	Purchases	Correct
Goods available for sale	Correct	Goods available for sale	Overstated
Ending inventory	Overstated	Ending inventory	Correct
Cost of goods sold	Understated	Cost of goods sold	Overstated
Gross profit	Overstated	Gross profit	Understated
Operating expenses	Correct	Operating expenses	Correct
Net income	Overstated	Net income	Understated
Balance Sheet		Balance Sheet	
Ending inventory	Overstated	Ending inventory	Correct
Current assets	Overstated	Current assets	Correct
Total assets	Overstated	Total assets	Correct
Retained earnings	Overstated	Retained earnings	Correct
Total stockholders' equity	Overstated	Total stockholders' equity	Correct
Total liabilities and stockholders' equity	Overstated	Total liabilities and stockholders' equity	Correct

Effect on net income reverses in next year

Effect on balance sheet reverses in next year

Ending Inventory **Understated** in Current Period

Current Period			Next Period		
Income Statement			Income Statement		
Sales		Correct	Sales		Correct
Beginning inventory	Correct		Beginning inventory	Understated	
Purchases	Correct		Purchases	Correct	
Goods available for sale	Correct		Goods available for sale	Understated	
Ending inventory	Understated		Ending inventory	Correct	
Cost of goods sold		Overstated	Cost of goods sold		Understated
Gross profit		Understated	Gross profit		Overstated
Operating expenses		Correct	Operating expenses		Correct
Net income		Understated	Net income		Overstated
Balance Sheet			Balance Sheet		
Ending inventory	Understated		Ending inventory	Correct	
Current assets	Understated		Current assets	Correct	
Total assets	Understated		Total assets	Correct	
Retained earnings	Understated		Retained earnings	Correct	
Total stockholders' equity	Understated		Total stockholders' equity	Correct	
Total liabilities and stockholders' equity	Understated		Total liabilities and stockholders' equity	Correct	

Effect on net income reverses in next year

Effect on balance sheet reverses in next year

To illustrate, assume that Zula Industries incorrectly counted its December 31, 20Y1, inventory at \$250,000 instead of the correct amount of \$220,000. The effect of the misstatement on Zula's income statement and balance sheet for 20Y1 (current year) and 20Y2 (following year) is shown in [Exhibit 12](#).

Exhibit 12

Effects of Inventory Errors—Zula Industries

Zula Industries Income Statement For the Year Ended December 31, 20Y1			Zula Industries Income Statement For the Year Ended December 31, 20Y2		
Sales		Correct	Sales		Correct
Beginning inventory	Correct		Beginning inventory	\$30,000 over	
Purchases	Correct		Purchases	Correct	
Goods available for sale	Correct		Goods available for sale	\$30,000 over	
Ending inventory	\$30,000 over *		Ending inventory	Correct	
Cost of goods sold		\$30,000 under	Cost of goods sold		\$30,000 over
Gross profit		\$30,000 over	Gross profit		\$30,000 under
Operating expenses		Correct	Operating expenses		Correct
Net income		\$30,000 over	Net income		\$30,000 under
Zula Industries Balance Sheet December 31, 20Y1			Zula Industries Balance Sheet December 31, 20Y2		
Ending inventory	\$30,000 over		Ending inventory	Correct	
Current assets	\$30,000 over		Current assets	Correct	
Total assets	\$30,000 over		Total assets	Correct	
Retained earnings	\$30,000 over		Retained earnings	Correct	
Total stockholders' equity	\$30,000 over		Total stockholders' equity	Correct	
Total liabilities and stockholders' equity	\$30,000 over		Total liabilities and stockholders' equity	Correct	

* \$250,000 - \$220,000 = \$30,000

Effect on net income reverses in next year

Effect on balance sheet reverses in next year

Check up Corner 6-5

Effects of Inventory Errors

Zulu Industries incorrectly counted its December 31, 20Y1, inventory at \$250,000 instead of the correct amount of \$220,000. Indicate the effect of the misstatement on Zulu's income statement for the current year (20Y1) and the following year (20Y2). What is the net effect of the error for the two years?

Analysis for Decision Making

Inventory Turnover and Days' Sales in Inventory

A merchandising business should keep enough inventory on hand to meet its customers' needs. A failure to do so may result in lost sales. However, too much inventory ties up funds that could be used to improve operations.

Also, excess inventory increases expenses such as storage and property taxes. Finally, excess inventory increases the risk of losses due to price decreases, damage, or changes in customer tastes.

Objective 7 - Describe and illustrate how inventory turnover and the days' sales in inventory are used to analyze the efficiency and effectiveness of inventory management.

Two measures to analyze inventory management are:

- Inventory turnover
- Days' sales in inventory

Inventory turnover (A measure of the number of times inventory is turned into goods sold during the year, computed by dividing the cost of goods sold by the average inventory.)

measures the relationship between the cost of goods sold and the amount of inventory carried during the period. It measures the number of times inventory is turned into sold goods during the year. It is computed as follows:

$$\text{Inventory Turnover} = \frac{\text{Cost of Goods Sold}}{\text{Average Inventory}}$$

To illustrate, inventory turnover for **Best Buy Co., Inc. (BBY)** is computed from the following data (in millions) from two recent annual reports:

	Year 3	Year 2	Year 1
Cost of goods sold	\$32,918	\$32,275	\$29,963

Inventories:

Beginning of year	\$5,209	\$4,864	\$5,051
End of year	\$5,409	\$5,209	\$4,864

Average inventory:

$(\$5,209 + \$5,409) \div 2$	\$5,309.0		
$(\$4,864 + \$5,209) \div 2$		\$5,036.5	
$(\$5,051 + \$4,864) \div 2$			\$4,957.5

Inventory turnover:

$\$32,918 \div \$5,309.0$	6.2		
$\$32,275 \div \$5,036.5$		6.4	
$\$29,963 \div \$4,957.5$			6.0

Generally, the larger the inventory turnover, the more efficient and effective the company is in managing inventory. In the preceding example, inventory turnover increased from 6.0 to 6.4 during Year 2; thus, Best Buy's inventory management improved during Year 2. Between Year 2 and Year 3, inventory turnover declined slightly from 6.4 to 6.2.

The

days' sales in inventory (The measure of the length of time it takes to acquire, sell, and replace inventory, computed by dividing the average inventory by the average daily cost of goods sold.)

measures the length of time it takes to acquire, sell, and replace the inventory. It is computed as follows:

$$\text{Days' Sales in Inventory} = \frac{\text{Average Inventory}}{\text{Average Daily Cost of Goods Sold}}$$

The average daily cost of goods sold is determined by dividing the cost of goods sold by 365. Based upon the preceding data, the days' sales in inventory for **Best Buy** is computed as follows:

	Year 3	Year 2	Year 1
Cost of goods sold	\$32,918	\$32,275	\$29,963
Average daily cost of goods sold:			
$\$32,918 \div 365 \text{ days}$	\$90.2 per day		
$\$32,275 \div 365 \text{ days}$		\$88.4 per day	

$\$29,963 \div 365 \text{ days}$			\$82.1 per day
Average inventory	\$5,309.0	\$5,036.5	\$4,957.5
Days' sales in inventory:			
$\$5,309.0 \div \90.2	58.9 days		
$\$5,036.5 \div \88.4		57.0 days	
$\$4,957.5 \div \82.1			60.4 days

Generally, the lower the days' sales in inventory, the more efficient and effective the company is in managing inventory. The days' sales in inventory decreased from 60.4 days in Year 1 to 58.9 days in Year 3; thus, Best Buy's inventory management improved. This is consistent with the increase in inventory turnover during the year.

As with most financial ratios, differences exist among industries. To illustrate, **The Kroger Co. (KR)** is the world's largest grocery store chain. Because food is perishable, it will sell more rapidly than Best Buy's consumer electronics. Thus, Kroger's inventory management should be significantly more efficient than Best Buy's. For a recent year, this is confirmed as follows:

	Best Buy	Kroger
Inventory turnover	6.2	14.2
Days' sales in inventory	58.9 days	25.7 days

Make a Decision

Inventory Turnover and Days' Sales in Inventory

Analyze and compare Amazon.com to Target (MAD 6-1) (Continuing company analysis)

Analyze and compare Darden Restaurants to Chipotle Mexican Grill (MAD 6-2)

Analyze and compare Costco, Walmart, and Nordstrom (MAD 6-3)

Analyze and compare Monster Beverage and Brown-Forman (MAD 6-4)

